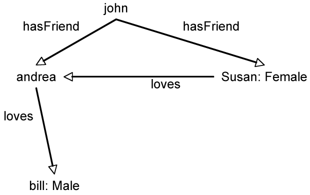
Exercise

Problem: Assume the following KB



$$Male = \neg Female$$

$(john, susan){:}hasFriend$
$(john, andrea){:}hasFriend$
$(susan, andrea){:}loves$
$(andrea, bill){:}loves$

$susan{:}Female$
$bill{:}Male$

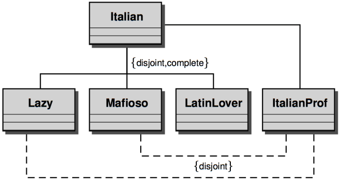
Is it true that “John has a female friend loving a male person”? that is, is it true that

$$\mathcal{K} \models john{:}\exists hasFriend.(Female \sqcap (\exists loves.Male))$$

Solution: Yes, because andrea is either male or female . . .

Exercise

Problem: Consider the following conceptual schema



Encode it into Description logics and prove that $\mathcal{K} \models \mathit{ItalianProf} \sqsubseteq \mathit{LatinLover}$ is implied (i.e. "an *ItalianProf* is a *LatinLover*")

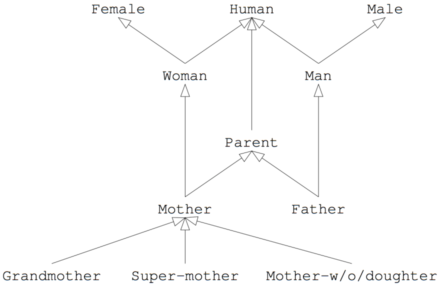
Solution:

$$
\begin{array}{rcl}
\mathit{Lazy} & \sqsubseteq & \mathit{Italian} \\
\mathit{Mafioso} & \sqsubseteq & \mathit{Italian} \\
\mathit{LatinLover} & \sqsubseteq & \mathit{Italian} \\
\mathit{ItalianProf} & \sqsubseteq & \mathit{Italian} \\
\mathit{Lazy} & \sqsubseteq & \neg \mathit{Mafioso} \\
\mathit{Lazy} & \sqsubseteq & \neg \mathit{LatinLover} \\
\mathit{Mafioso} & \sqsubseteq & \neg \mathit{LatinLover} \\
\mathit{Mafioso} & \sqsubseteq & \neg \mathit{ItalianProf} \\
\mathit{Lazy} & \sqsubseteq & \neg \mathit{ItalianProf}
\end{array}
$$

An *ItalianProf* is Italian, hence has to be either *Lazy*, *Mafioso* or *Latinlover*. But, it cannot be the former two, so it has to be last one, i.e. a *LatinLover*.

Exercise

Problem: Consider the following conceptual schema



Encode it into Description logics

Solution:

$$
\begin{array}{rcl}
Grandmother & \sqsubseteq & Mother \\
SuperMother & \sqsubseteq & Mother \\
MotherWODaughter & \sqsubseteq & Mother \\
Mother & \sqsubseteq & Parent \sqcap Woman \\
Father & \sqsubseteq & Parent \sqcap Man \\
Parent & \sqsubseteq & Human \\
Woman & \sqsubseteq & Femal \sqcap Human \\
Man & \sqsubseteq & Human \sqcap Male
\end{array}
$$

Exercise

Problem: Assume the following KB

$$\exists teaches.Course \sqsubseteq (Student \sqcap \exists degree.BS) \sqcup Prof$$
$$Prof \sqsubseteq \exists degree.MS$$
$$\exists degree.MS \sqsubseteq \exists degree.BS$$
$$MS \sqcap BS \sqsubseteq \bot$$

$$(john, sistemiIntelligentiI):teaches$$
$$john:(\leq 1\ degree)$$
$$sistemiIntelligentiI:Course$$

Is it true that "John is a student" ? That is, is it true that

$$\mathcal{K} \models john:Student\ ?$$

Solution: Yes, because

1. $john:\exists teaches.Course$ and, thus, either $john:(Student \sqcap \exists degree.BS)$ or $john:Prof$
2. if $john:Prof$ then $john:\exists degree.MS$ and, thus, $john:\exists degree.BS$
3. but, $MS \sqcap BS \sqsubseteq \bot$, i.e. MS and BS are disjoint and, thus $john:(\geq 2\ degree)$ (john has at least two degrees)
4. but, $john:(\leq 1\ degree) \in \mathcal{K}$ (i.e., john has at most one degree), a constradiction, and, thus, it cannot be that $john:Prof$
5. Therefore, $john:(Student \sqcap \exists degree.BS)$ is true and, thus, $john:Student$.

Exercise

Problem: Transform $\neg(A \sqcap \neg B \sqcap (\exists R.(C \sqcup \neg D \sqcup (\forall P.\neg D))))$ into Negation Normal Form

Solution: $\neg A \sqcup B \sqcup (\forall R.(\neg C \sqcap D \sqcap (\exists P.D)))$